KERALA PUBLIC SERVICE COMMISSION • OFFICIAL SYLLABUS NOTE

SURVEYOR GRADE II (SURVEY & LAND RECORDS) & KWA TRACER

Cat. Nos: 411/2022, 692/2022, 106/2017, 043/2023 • Technical Education & Town Planning

MODULE 08

Computer Aided Drafting (AutoCAD)

Official Weightage: 10 Marks Marks

PRESCRIBED SYLLABUS TOPICS COVERED:

CAD Concepts & Interface (GUI, Application menu, Ribbon, Command line, File formats .dwg, .dxf, .dwt, .bak), Coordinate Systems (WCS, UCS, Absolute, Relative Rectangular, Relative Polar), Drawing Setup (UNITS, LIMITS, ZOOM, PAN, REGEN), Draw Commands (Line, Polyline, Circle, Arc, Rectangle, Polygon, Ellipse, Hatch, Spline), Modify Commands (Move, Copy, Rotate, Scale, Trim, Extend, Offset, Mirror, Fillet, Chamfer, Array, Explode, Stretch), Drafting Aids & Function Keys (F1 to F12, OSNAP, Ortho, Polar tracking), Layers & Properties (Layer states, Defpoints, Linetypes, LTSCALE), Dimensioning & Text (DIMSTYLE, MTEXT, DTEXT), Blocks & Attributes (BLOCK, WBLOCK, ATTDEF), Layouts & Plotting.

+P COMPREHENSIVE EXAM CRACKER FEATURES

- Strictly aligned with Kerala PSC latest question patterns (Land Records & KWA)
- Contains verified formulas, numerical shortcuts, standard tables & specifications
- Highlights repeatedly tested PSC trap questions & previous year keys

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CHAPTER 1: INTRODUCTION TO CAD & AUTOCAD ENVIRONMENT

Computer Aided Drafting (CAD) is the use of computer software to create, modify, analyze, and optimize technical drawings. In Kerala PSC engineering exams (Surveyor Gr. II, KWA Tracer, LSGD Draftsman), AutoCAD commands, keyboard shortcuts, coordinate formats, and drafting aids are heavily tested.

1.1 AutoCAD User Interface (GUI) Components

• **Application Menu (Big 'A'):** Located at the upper-left corner of the window. Provides access to file management tools: New, Open, Save, Save As, Export, Print/Plot, and Close.

• **Quick Access Toolbar (QAT):** Displays frequently used commands: New, Open, Save, Undo (Ctrl+Z), Redo (Ctrl+Y), Plot (Ctrl+P), and Workspace Switching.

• **Ribbon & Panels:** Organizes commands into logical Tabs (Home, Insert, Annotate, Parametric, View, Manage, Output) containing Panels (Draw, Modify, Annotation, Layers, Block, Properties, Utilities).

• **Command Line Window:** The heart of AutoCAD. Located at the bottom of the screen. Displays command prompts, options in square brackets [ ], default values in angled brackets < >, and accepts typed command shortcuts.

• **Status Bar:** Located along the bottom edge of the screen. Contains toggle icons for drafting aids: Grid (F7), Snap (F9), Ortho (F8), Polar Tracking (F10), OSNAP (F3), Object Snap Tracking (F11), Dynamic Input (F12), and Workspace Switching.

• **Crosshair Cursor:** Composed of two intersecting perpendicular lines with a small square box at the center called the PICKBOX (used for selecting objects).

1.2 Standard AutoCAD File Formats & Extensions

File Extension	Full Technical Designation	Description & Standard Function in CAD
.dwg	Drawing Database File	AutoCAD's native, proprietary binary vector drawing format; stores all 2D and 3D geometric entities, layers, and metadata
.dxf	Drawing Exchange Format	Universal ASCII or binary CAD interchange format developed by Autodesk to enable interoperability with GIS, CAM, and other CAD programs
.dwt	Drawing Template File	A prototype drawing file containing predefined layers, text styles, dimension styles, units, and title blocks (e.g. acad.dwt, acadiso.dwt)
.bak	Backup Drawing File	Automatic backup created in the same folder whenever an existing .dwg is saved; preserves the state of the drawing PRIOR to the latest save; rename to .dwg to recover
.sv$	Automatic Save File	Temporary file created at user-defined time intervals (default 10 mins) stored in system Temp folder; used for emergency crash recovery

+P PSC FOCUS POINT: BAK FILE RECOVERY IN PSC

PSC Question: 'How can you recover a drawing from a .bak file in AutoCAD?' Rename the file extension from .bak to .dwg in Windows File Explorer! The .bak file is created automatically every time a drawing is saved.

CHAPTER 2: COORDINATE SYSTEMS & INPUT MODES

AutoCAD uses the Cartesian coordinate system to locate points in 2D and 3D space. The intersection of the X-axis and Y-axis is the Origin (0,0).

2.1 World Coordinate System (WCS) vs User Coordinate System (UCS)

• **World Coordinate System (WCS):** The fixed, permanent global reference coordinate system. The X-axis is horizontal (positive to the right), Y-axis is vertical (positive upwards), and Z-axis is perpendicular to screen (positive towards

observer). WCS cannot be altered.

• **User Coordinate System (UCS):** A movable, user-defined Cartesian coordinate system that can be positioned, rotated, and inclined anywhere in 3D space to simplify drafting on inclined planes or local building alignments. Commanded by UCS.

2.2 The Three Coordinate Entry Methods

Coordinate Method	Syntax / Format	Example Entry	Mathematical Reference / Explanation
Absolute Coordinates	X, Y	50, 75	Locates a point strictly with reference to the fixed Origin (0,0). X = 50 units right of origin, Y = 75 units above origin
Relative Rectangular Coordinates	@9E,Â ”Y	@30, 40	Locates a point relative to the LAST ENTERED POINT preceded by '@'. Moves 30 units in X and 40 units in Y from current point
Relative Polar Coordinates	@Distance < Angle	@100 < 45	Locates a point at a specified linear distance and angle from the LAST POINT. Distance = 100 units at an angle of 45° counter-clockwise from positive X-axis
Direct Distance Entry (DDE)	Distance + Direction	Type '100' + Enter	Pointer is aimed in the desired direction (locked with Ortho F8 or Polar Tracking F10) and exact distance is typed directly

+P PSC FOCUS POINT: ANGULAR CONVENTION IN AUTOCAD

In standard AutoCAD default settings:

• 0° is along the positive X-axis (pointing East / Right).

• Angles are measured POSITIVE in the COUNTER-CLOCKWISE direction.

• 90° = North (Up), 180° = West (Left), 270° = South (Down).

CHAPTER 3: DRAWING SETUP & DISPLAY CONTROLS

Setting up the workspace units, sheet limits, and display magnification ensures correct scale and visual precision.

3.1 Drawing Setup Commands: UNITS & LIMITS

• **UNITS Command (UN):** Controls display format and precision of linear lengths and angular measurements. Length types: Architectural (feet & inches), Decimal (metric mm/m), Engineering (feet & decimal inches), Fractional, Scientific. Angle types: Decimal Degrees, Deg/Min/Sec, Grads, Radians, Surveyor's Units.

• **LIMITS Command:** Sets the artificial boundaries of the drawing grid. Entered as lower-left corner (usually 0,0) and upper-right corner (e.g. 420,297 for A3 sheet). LIMITS ON prevents drawing outside bounds.

3.2 Display Controls: ZOOM, PAN & REGEN

Command / Option	Shortcut	Function & Operational Characteristics
ZOOM All	Z -> A	Zooms to display the entire drawing area defined by the drawing LIMITS or the drawing extents, whichever is larger
ZOOM Extents	Z -> E	Zooms to display ALL DRAWN OBJECTS in the drawing to fill the current graphics screen area (independent of drawing limits)
ZOOM Window	Z -> W	Zooms to display an area specified by two diagonally opposite pick points
ZOOM Previous	Z -> P	Restores the previous display view (up to 10 previous views)

Command / Option	Shortcut	Function & Operational Characteristics
PAN	P	Shifts the display view in any direction without changing the magnification scale factor
REGEN	RE	Regenerates the entire drawing from floating-point database; recalculates screen coordinates and restores smooth curves
VIEWRES	VIEWRES	Sets the resolution for fast zoom and controls circle smoothness (number of line segments drawn for circles)

CHAPTER 4: BASIC DRAWING (DRAW) COMMANDS

Draw commands create primary 2D vector geometric entities in the AutoCAD drawing database.

Command Name	Alias	Key Sub-Options & Specifications	PSC Exam Key Points
LINE	L	Specify first point -> specify next point. Closed by typing 'C'	Each line segment is an INDEPENDENT separate object
POLYLINE	PL	Arc, Halfwidth, Length, Undo, Width	Creates a single continuous connected 2D object consisting of line and arc segments; can have variable line widths!
CIRCLE	C	Center-Radius (default), Center-Diameter, 2P (two points), 3P (three points), TTR (Tangent-Tangent-Radius), TTT	Standard default mode is Center-Radius
ARC	A	11 different methods: 3-Point (default), Start-Center-End, Start-Center-Angle, Start-Center-Length, Start-End-Radius	AutoCAD draws arcs in COUNTER-CLOCKWISE direction by default!
RECTANGLE	REC	Chamfer, Elevation, Fillet, Thickness, Width, Area, Dimensions, Rotation	Creates a closed POLYLINE object with 4 vertices
POLYGON	POL	Number of sides (3 to 1024), Inscribed in circle (I), Circumscribed about circle (C), Edge (E)	Creates a regular closed equilateral polyline
ELLIPSE	EL	Axis endpoints (Major and minor axes), Center, Elliptical Arc	Mathematical true ellipse
SPLINE	SPL	Fit points, Control Vertices (CV), Degree, Knot parameterization	Smooth NURBS curve passing through or guided by points
HATCH	H	Pick internal point, Select boundary objects, Pattern (ANSI, ISO, Solid, Gradient), Scale, Angle, Associative	Associative hatch automatically updates shape when boundary is modified!

& þ COMMON PSC EXAM TRAP: LINE VS POLYLINE IN PSC

• LINE command creates individual, disconnected line segments.
• POLYLINE (PL) creates a SINGLE unified compound object comprising multiple straight and curved segments with customizable line width.
• To convert Lines into a Polyline, use PEDIT (PE) with the 'Join' option.

CHAPTER 5: OBJECT MODIFICATION (MODIFY) COMMANDS

Modify commands alter the geometry, position, size, and orientation of existing drawing objects.

Command	Alias	Operational Mechanism	PSC Key Behavior

Command	Alias	Operational Mechanism	PSC Key Behavior
ERASE	E	Deletes selected objects from the drawing database	OOPS command restores the most recently erased objects without undoing intervening work!
MOVE	M	Displaces objects by a specified distance and direction from a base point	Original object shifts to new position; source deleted
COPY	CO / CP	Duplicates objects by a specified displacement vector from base point	Original remains intact; multiple copies can be placed
ROTATE	RO	Rotates objects about a specified base point by an angle	Counter-clockwise angle is positive; 'Reference' option matches existing alignments
SCALE	SC	Enlarges or reduces objects proportionally about a base point	Scale factor > 1 enlarges; Scale factor < 1 reduces (e.g. 0.5 halves size); 'Reference' option available
OFFSET	O	Constructs parallel lines, concentric circles, and parallel curves	Specify offset distance -> select object -> click side to offset
TRIM	TR	Cuts objects cleanly at edges defined by other intersecting objects	Shift+Select toggles to EXTEND command!
EXTEND	EX	Lengthens objects to meet boundary edges of other objects	Shift+Select toggles to TRIM command!
FILLET	F	Connects two intersecting or non-parallel lines with a smooth tangent arc of radius R	Setting Fillet Radius R = 0 creates a clean sharp corner!
CHAMFER	CHA	Bevels the intersection of two lines with an inclined straight segment	Defined by two distances (D1, D2) or distance and angle
MIRROR	MI	Creates a reverse reflection copy of objects across a specified mirror line	Controlled by MIRRTEXT system variable (0 = text not reversed; 1 = text reversed)
ARRAY	AR	Creates multiple copies in organized Rectangular, Polar (circular), or Path patterns	Associative arrays remain linked and dynamically editable
EXPLODE	X	Decomposes compound entities (polylines, blocks, dimensions, hatches, meshes) into component lines and arcs	Cannot explode a non-compound entity; Mtext explodes into single-line text

+P PSC FOCUS POINT: FILLET RADIUS ZERO & OOPS COMMAND

• Setting Fillet Radius R = 0 trims or extends two non-parallel lines to form a perfect sharp 90° corner.

• OOPS command restores the last deleted selection set even after performing other intermediate commands (unlike Undo which reverts everything).

CHAPTER 6: DRAFTING AIDS, FUNCTION KEYS & OBJECT SNAPS

Drafting aids allow surveyors to draft with mathematical precision without calculating coordinates manually.

6.1 Function Keys Quick Reference Table

Function Key	Drafting Aid Feature	Operational Toggle Function
F1	AutoCAD Online Help	Opens AutoCAD documentation and command reference
F2	Expanded Text History Window	Toggles command line text history window on/off
F3	Object Snap (OSNAP)	Toggles running object snap modes on/off

Function Key	Drafting Aid Feature	Operational Toggle Function
F4	3D Object Snap	Toggles 3D object snap modes on/off
F5	Isoplane Switching	Cycles isometric planes: Isoplane Left, Top, Right
F6	Dynamic UCS	Toggles automatic dynamic user coordinate system on 3D faces
F7	Grid Display	Toggles graphics display background grid on/off
F8	Ortho Mode	Locks cursor movement strictly to horizontal and vertical axes (0°, 90°, 180°, 270°)
F9	Snap Mode	Locks crosshair movement to discrete grid snap increments
F10	Polar Tracking	Displays alignment paths at specified polar angle increments (15°, 30°, 45°, 90°)
F11	Object Snap Tracking (OTRACK)	Tracks along alignment paths from active OSNAP points
F12	Dynamic Input	Displays command prompts and coordinate inputs directly beside crosshairs

6.2 Object Snap (OSNAP) Modes & Selection Modes

• **OSNAP Modes:** ENDpoint (snaps to line/arc ends), MIDpoint (snaps to exact midpoint of segment), CENter (snaps to center of circle/arc/ellipse), NODe (snaps to point object), QUADrant (snaps to 0°, 90°, 180°, 270° circle points), INTersection (snaps to intersection of two objects), EXTension (tracks along imaginary extension line), PERpendicular (snaps to perpendicular drop point), TANgent (snaps to circle/arc tangent), NEArest (snaps to nearest point on object).

• **Window Selection (Blue Box):** Drawn from LEFT to RIGHT. Selects ONLY objects that lie ENTIRELY inside the rectangular selection box.

• **Crossing Selection (Green Box):** Drawn from RIGHT to LEFT with dashed boundary. Selects all objects that lie inside OR are TOUCHED/CROSSED by the selection box.

& þ COMMON PSC EXAM TRAP: WINDOW VS CROSSING SELECTION IN PSC

• Left-to-Right = Solid Blue window => selects ONLY objects completely enclosed.

• Right-to-Left = Dashed Green window => selects ANY object touched or crossed by the box!

CHAPTER 7: LAYERS, PROPERTIES & DRAWING STANDARDS

Layers are the primary organizational tool in AutoCAD, functioning like transparent electronic overlays that organize drawing elements by functional category (e.g. Centerlines, Wall Outlines, Dimensions, Text, Electrical).

7.1 Layer States and Controls

Layer State	Icon	Visual Display	Database Regeneration & Performance Effect
On / Off	Lightbulb	Turning OFF makes objects invisible; turning ON makes objects visible	Objects are still evaluated in memory during REGEN (drawings still calculate frozen geometry)
Freeze / Thaw	Sun / Snowflake	Freezing makes objects completely invisible; thawing restores visibility	Frozen layers are NOT calculated or regenerated in memory; drastically improves regeneration speed in large civil master plans!

Layer State	Icon	Visual Display	Database Regeneration & Performance Effect
Lock / Unlock	Padlock	Locked layer objects remain fully visible but cannot be edited, moved, or deleted by accident	Prevents accidental erasure; objects can still be snapped to using OSNAP!
Plot / No-Plot	Printer	Controls whether the layer will be printed/plotted	No-plot layers appear on screen for reference but do NOT print on final paper plot (e.g. viewports)

+P PSC FOCUS POINT: LAYER 0 & DEFPOINTS SPECIAL PROPERTIES

• Layer 0: The default system layer present in every drawing; CANNOT be deleted or renamed.

• Defpoints Layer: Created automatically when dimensions are added; objects placed on Defpoints are visible on screen but NEVER PLOT on paper plots!

7.2 Linetype Scale & Match Properties

• **LTSCALE Command:** Sets the global linetype scale factor for the entire drawing. Controls the spacing and frequency of dashes and dots in hidden and centerlines.

• **MATCHPROP Command (MA):** Copies the properties (layer, color, linetype, lineweight, dimension style, hatch pattern) from a source object to one or more destination objects.

CHAPTER 8: ANNOTATION, BLOCKS & PLOTTING

Annotations communicate manufacturing, dimensioning, and cadastral survey information to the reader.

8.1 Text: Single-Line (DTEXT) vs Multiline (MTEXT)

• **Single-Line Text (DTEXT / TEXT):** Each line of text is created as an independent object. Ideal for short survey labels, benchmark notes, room names.

• **Multiline Text (MTEXT):** Creates a single unified paragraph text object with a rich text formatting editor (fonts, columns, bullets, stacked fractions).

• **Special Text Symbols:** %%c = Diameter symbol (Ø); %%d = Degree symbol (°); %%p = Plus/Minus tolerance symbol (±); %%% = Percentage symbol (%).

8.2 Blocks & Attributes: BLOCK vs WBLOCK

Block Command	Command Alias	Storage Scope & Nature	PSC Examination Key
BLOCK	B	Creates an internal block definition stored strictly WITHIN the current drawing database	Available only in the drawing where it was created
WBLOCK (Write Block)	W	Exports a block or selected geometry as an INDEPENDENT external .dwg file saved to disk	Can be inserted into ANY other drawing as a library component
ATTDEF	ATT	Creates an Attribute: an intelligent, editable text label embedded inside a block definition	Allows prompt-driven data entry (e.g. Survey Station ID, Bench Mark RL) when block is inserted

8.3 Model Space vs Paper Space & Viewports

• **Model Space:** The infinite, 3D design workspace where the actual engineering structure or land parcel is drawn at TRUE FULL SIZE (1:1 scale).

• **Paper Space (Layout Tabs):** The 2D page-layout environment representing the physical sheet of paper where drawings are arranged, scaled, and bordered for final printing.

• **Floating Viewports (MVIEW / MV):** Windows cut into paper space through which model space objects are viewed. Allows setting different display scales (e.g. 1:500 for site plan, 1:50 for detail section) on the same sheet!

8.4 Basic 3D Modeling Commands in AutoCAD

3D Command	Alias	3D Geometric Operation	Standard Engineering Application
EXTRUDE	EXT	Creates a 3D solid or surface by extruding a 2D closed planar curve (profile) along a path or perpendicular distance	Extruding building floor plans into 3D walls, pillars, bridge piers
REVOLVE	REV	Sweeps a 2D closed profile around a specified revolution axis through a given angle (default 360°)	Creating circular symmetrical solids: pipes, domes, shafts, pulleys, tanks
SWEEP	SWEEP	Extrudes a 2D closed profile along an open or closed 2D/3D path curve	Water supply pipes, handrails, road alignments, ductwork
LOFT	LOFT	Creates a 3D solid by creating transitions through a series of cross-sectional profiles	Boat hulls, airplane wings, complex transition topography
UNION	UNI	Combines two or more overlapping 3D solids or 2D regions into a single unified composite solid	Boolean addition of solid objects
SUBTRACT	SU	Removes the volume of one 3D solid from another (base solid - cutting solid)	Cutting holes, windows, and door openings in solid 3D walls
INTERSECT	IN	Creates a 3D solid from the common overlapping volume of two or more intersecting solids	Boolean intersection operations

CHAPTER 9: COMPLETE AUTOCAD SHORTCUT KEYS QUICK REFERENCE TABLE

Shortcut	Command Name	Shortcut	Command Name	Shortcut	Command Name
L	LINE	M	MOVE	LA	LAYER
PL	POLYLINE	CO / CP	COPY	LT	LINETYPE
C	CIRCLE	RO	ROTATE	LTS	LTSCALE
A	ARC	SC	SCALE	MA	MATCHPROP
REC	RECTANGLE	TR	TRIM	B	BLOCK
POL	POLYGON	EX	EXTEND	W	WBLOCK
EL	ELLIPSE	F	FILLET	I	INSERT
H	HATCH	CHA	CHAMFER	MT	MTEXT
REG	REGION	O	OFFSET	D	DIMSTYLE
SPL	SPLINE	MI	MIRROR	Z	ZOOM
X	EXPLODE	AR	ARRAY	P	PAN

CHAPTER 10: KERALA PSC 50 RAPID-FIRE ONE-LINERS & EXAM VAULT

Rapid-Fire PSC One-Liners: GUI & Coordinate Systems

• **Native proprietary file extension of AutoCAD:** .dwg (Drawing Database).

• **Universal CAD interchange file format:** .dxf (Drawing Exchange Format).

• **File format for AutoCAD drawing templates:** .dwt (Drawing Template).

- **Automatic backup file extension:** .bak (Backup Drawing).
- **Symbol prefix for relative coordinates:** @ symbol (e.g. @50,30).
- **Default angle measurement direction in AutoCAD:** Counter-Clockwise (Positive angles).
- **Coordinate system with fixed origin (0,0,0):** World Coordinate System (WCS).
- **User-defined movable coordinate system:** User Coordinate System (UCS).
- **Command to set drawing sheet boundaries:** LIMITS.
- **Command to restore smooth circles after zooming:** REGEN (RE).

Rapid-Fire PSC One-Liners: Commands & Function Keys

- **Function key to toggle Ortho mode on/off:** F8.
- **Function key to toggle OSNAP on/off:** F3.
- **Function key to toggle Grid on/off:** F7.
- **Function key to toggle Polar Tracking:** F10.
- **Function key to toggle Dynamic Input:** F12.
- **Command to create rounded corner with arc:** FILLET (F).
- **Fillet radius value to create sharp 90° corner:** Radius = 0.
- **Command to bevel corners with straight segment:** CHAMFER (CHA).
- **Command to break polyline into individual lines:** EXPLODE (X).
- **Command to create parallel lines at given distance:** OFFSET (O).

Rapid-Fire PSC One-Liners: Layers, Blocks & Annotation

- **Layer that cannot be deleted or renamed:** Layer 0.
- **Layer whose objects NEVER plot on paper:** Defpoints layer.
- **Layer state that skips memory calculation in REGEN:** Freeze (Thaw).
- **Text code to write Diameter symbol (Ø):** %%c.
- **Text code to write Degree symbol (°):** %%d.
- **Text code to write Plus/Minus symbol (±):** %%p.
- **Command to create external block file on disk:** WBLOCK (Write Block).
- **Command to create editable text inside a block:** ATTDEF (Attribute Definition).
- **Space where actual structure is drawn at 1:1 scale:** Model Space.
- **Command to create floating viewport in layout:** MVIEW (MV).

AutoCAD Drafting Revision Checklist Matrix

Topic / Command	Governing Rule / Syntax	Kerala PSC Exam Key
Coordinate Entry	Absolute: X,Y; Relative: @X,Y; Polar: @D<A	Counter-clockwise angle is positive; 0° is East
Function Keys	F3 = OSNAP; F8 = ORTHO; F10 = Polar; F7 = Grid	F8 locks to 90° increments; F3 toggles snaps
Line vs Polyline	Line = independent; Polyline = single object	Polyline can have variable width and curved arcs
Fillet Radius 0	F -> R -> 0	Creates sharp 90° corner between lines
Special Symbols	%%c = Ø, %%d = °, %%p = ±	Used in MTEXT and DTEXT annotation
Block vs Wblock	B = internal drawing only; W = external .dwg file	Wblock can be reused across multiple projects
Layer States	Freeze skips REGEN calculation; Off does not	Layer 0 cannot be deleted; Defpoints does not plot